BP101T COMPANION TEXT

# Python for Pharmacists

Unit V · Data Visualization with Matplotlib

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Course BP101T · B.Pharm Semester I

Chapters 15–17 · Six teaching hours

Theory-first · Line, histogram, scatter & box plots · concentration-time curves, ADR rates, dissolution profiles

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**Unit hours:** 6 · **Book chapters:** 15–17 · **Syllabus bullets covered:** introduction to Matplotlib; creating line plots, histograms, scatter plots, and box plots; labeling axes, titles, and legends; creating plots and visualising pharmaceutical datasets — concentration-time curves for oral and IV administration, ADR reporting rates across drugs, dissolution profiles; scientific interpretation of plots.

**Unit objectives.** After studying this unit you should be able to: (1) import and use Matplotlib's pyplot interface; (2) create the four core plot types — line, histogram, scatter and box plot; (3) label axes, add titles and display legends; (4) plot pharmaceutical datasets including concentration-time curves, ADR rates and dissolution profiles; and (5) read and interpret a scientific plot correctly.

# Chapter 15 — Matplotlib Fundamentals

## 15.1 What Matplotlib is and why pharmacists plot

**Matplotlib** is Python'leading plotting library — converting raw numeric columns into figures suitable for journals and reports. In pharmaceutical research, a graph often reveals what raw numbers hide: the peak and fall of a concentration-time curve, the spread of dissolution results across batches, the dose at which a response plateaus. Visuals are not decoration; they are evidence, and regulatory submissions, research papers and hospital reports all demand them.

Install with `pip install matplotlib`, then import the pyplot interface with its universal alias:

```
import matplotlib.pyplot as plt
```

## 15.2 The plotting workflow

Every Matplotlib figure follows the same five steps:

1. **Pick the right chart** and pass the data (`plt.plot`, `plt.hist`, `plt.scatter`, `plt.boxplot`, `plt.bar`)
2. **Label the x-axis** (`plt.xlabel`)
3. **Label the y-axis** (`plt.ylabel`)
4. **Add a title** (`plt.title`)
5. **Display** (`plt.show`)

In a Jupyter notebook, `plt.show()` is optional — the figure renders below the cell automatically. In a script, it opens the display window.

## 15.3 Line plots — the workhorse

A line plot connects data points in order — the natural choice for time-series data such as concentration-time curves:

```
import matplotlib.pyplot as plt

time_h = [0, 2, 4, 6, 8, 10, 12]
conc = [0, 18, 25, 22, 16, 10, 6]

plt.plot(time_h, conc, color="teal", marker="o", linewidth=2)
plt.xlabel("Time (hours)")
plt.ylabel("Plasma Concentration (mcg/mL)")
plt.title("Drug Concentration vs Time")
plt.show()
```

This produces a smooth line with circular markers at each data point, teal colour, and fully labelled axes — the minimum standard for any scientific figure.

**Line-customisation parameters:**

| Parameter | Purpose | Example |
|---|---|---|
| `color` | line colour | `"teal"`, `"red"`, `"#0F6E6E"` |
| `marker` | point style at each datum | `"o"` circle, `"s"` square, `"^"` triangle |
| `linewidth` / `lw` | thickness of the line | `2` |
| `linestyle` / `ls` | line pattern | `"-"` solid, `"--"` dashed, `":"` dotted |
| `alpha` | transparency (0–1) | `0.7` |
| `label` | name for the legend | `"Drug A"` |

*Five line-plot examples:*

```
# 1. Simple line
plt.plot([1, 2, 3, 4], [10, 20, 25, 30])
```

```
# 2. With markers and colour
plt.plot(time_h, conc, color="teal", marker="o")
```

```
# 3. Dashed line for a reference curve
plt.plot(time_h, conc, linestyle="--", color="gray", label="Reference")
```

```
# 4. Two drugs overlaid
plt.plot(time_h, conc, color="teal", label="Drug A")
plt.plot(time_h, [c * 0.6 for c in conc], color="orange", label="Drug B")
plt.legend()
```

```
# 5. With a light grid
plt.plot(time_h, conc)
plt.grid(alpha=0.3)
```

The `label` parameter stores a name; `plt.legend()` displays a box matching each coloured line to its label — essential whenever more than one dataset appears on the same plot.

## 15.4 Histograms — the distribution view

A **histogram** shows how a single numeric variable is distributed by grouping values into **bins** and counting how many fall in each — the standard view for patient ages, doses or laboratory values.

```
ages = [23, 45, 34, 67, 45, 52, 29, 61, 38, 44, 55, 48]

plt.hist(ages, bins=6, color="orange", edgecolor="white")
plt.xlabel("Patient Age (years)")
plt.ylabel("Number of Patients")
plt.title("Distribution of Patient Ages")
plt.show()
```

**Choosing bins.** The `bins` setting determines into how many buckets values are grouped. Sparse buckets blur the shape; excessive buckets add clutter. Begin near `bins=10` and tune until the distribution reads clearly — adjust until the shape is clear without being jagged.

*More histogram examples:*

```
# Narrower bins (more detail)
plt.hist(ages, bins=12, color="teal", edgecolor="white")
```

```
# Wider bins (smoother)
plt.hist(ages, bins=4, color="orange", edgecolor="white")
```

```
# Dose distribution
doses = [250, 500, 500, 750, 500, 250, 1000, 500, 750, 500]
plt.hist(doses, bins=5, color="teal", edgecolor="white")
plt.xlabel("Dose (mg)")
plt.ylabel("Count")
```

```
# Cmax values across patients
cmax_values = [45.2, 38.6, 50.1, 42.8, 47.5, 39.2, 44.0, 41.3]
plt.hist(cmax_values, bins=8, color="orange", edgecolor="gray")
plt.xlabel("Cmax (mcg/mL)")
```

## 15.5 Scatter plots — the relationship view

A **scatter plot** draws individual data points for two variables on the same figure, exposing any association between the two variables — for example, dose versus therapeutic response.

```
dose = [50, 100, 150, 200, 250, 300, 350, 400]
response = [8, 18, 25, 33, 38, 44, 50, 55]

plt.scatter(dose, response, color="teal", alpha=0.75, s=60)
plt.xlabel("Dose (mg)")
plt.ylabel("Response (% reduction in symptom score)")
plt.title("Dose vs Therapeutic Response")
plt.show()
```

**Scatter customisation:**

| Parameter | Purpose |
|---|---|
| `s` | marker (dot) size |
| `alpha` | transparency (0 = invisible, 1 = solid) — helps when points overlap |
| `c` / `color` | point colour |

*More scatter examples:*

```
# Larger, semi-transparent points
plt.scatter(dose, response, s=100, alpha=0.5, color="teal")
```

```
# Age vs Cmax (looking for a relationship)
age = [34, 58, 45, 28, 62, 51, 40, 55]
cmax = [45.2, 38.6, 50.1, 42.8, 47.5, 39.2, 44.0, 41.3]
plt.scatter(age, cmax, color="orange", s=70)
plt.xlabel("Age (years)")
plt.ylabel("Cmax (mcg/mL)")
```

```
# Two groups in different colours
plt.scatter(dose[:4], response[:4], color="teal", label="Group A")
plt.scatter(dose[4:], response[4:], color="orange", label="Group B")
plt.legend()
```

## 15.6 Box plots — the five-number summary

A **box plot** (box-and-whisker) condenses a dataset into median, quartile and outlier markers — the natural choice when contrasting Cmax or dissolution results between several batches on one figure.

```
batch_a = [50, 55, 48, 52, 58, 47, 53]   # fast-release formulation
batch_b = [39, 41, 40, 38, 42, 40, 41]   # sustained-release
batch_c = [42, 47, 43, 45, 44, 46, 44]   # reference standard

plt.boxplot([batch_a, batch_b, batch_c],
            tick_labels=["Batch A", "Batch B", "Batch C"])
plt.ylabel("Cmax (mcg/mL)")
plt.title("Cmax Spread Across Three Batches")
plt.show()
```

**Anatomy of a box plot:**

| Part | Meaning |
|---|---|
| Box (top edge) | 75th percentile (Q3) |
| Line inside the box | median (50th percentile) |
| Box (bottom edge) | 25th percentile (Q1) |
| Whiskers | range of typical (non-outlier) values |
| Dots beyond whiskers | outliers — unusually high or low values |

A wide box signals significant manufacturing inconsistency (a quality flag); a tight box indicates consistent production.

*More box-plot examples:*

```
# Single batch
plt.boxplot(batch_a)
plt.ylabel("Cmax (mcg/mL)")
```

```
# Two batches, different colours
bp = plt.boxplot([batch_a, batch_b], patch_artist=True,
                 tick_labels=["Fast-release", "Sustained"])
bp["boxes"][0].set_facecolor("lightblue")
bp["boxes"][1].set_facecolor("lightyellow")
```

```
# Horizontal box plots (rotate for long labels)
plt.boxplot([batch_a, batch_b, batch_c], vert=False,
            tick_labels=["Formulation A", "Formulation B", "Reference"])
```

## 15.7 Labelling axes, titles and legends (syllabus point)

Every scientific plot must be fully labelled — axes without names are rejected by journals and examiners alike. When two or more datasets share the same axes, a legend is mandatory.

| Function | Purpose |
|---|---|
| `plt.xlabel("...")` | names the x-axis |

| Function | Purpose |
|---|---|
| `plt.ylabel("...")` | names the y-axis |
| `plt.title("...")` | adds a title above the plot |
| `label="..."` (in plot call) | sets the name shown in the legend |
| `plt.legend()` | displays the legend box; `loc` sets position (e.g. `"upper right"`) |
| `plt.grid(alpha=0.3)` | adds light gridlines for easier reading |

**The fully labelled standard** — every plot in this unit and beyond should look like this:

```
plt.plot(time_h, conc, color="teal", marker="o", label="Drug A")
plt.plot(time_h, [c * 0.6 for c in conc], color="orange",
         linestyle="--", label="Drug B")
plt.xlabel("Time (hours)")
plt.ylabel("Plasma Concentration (mcg/mL)")
plt.title("Comparing Two Drugs")
plt.legend(loc="upper right")
plt.grid(alpha=0.3)
plt.show()
```

**Chapter summary.** `import matplotlib.pyplot as plt`; the five-step workflow (plot → xlabel → ylabel → title → show). Line plots for time-series (`color`, `marker`, `linewidth`, `linestyle`, `label`); histograms for distributions (`bins`, `edgecolor`); scatter plots for relationships (`s`, `alpha`, `c`); box plots for five-number summaries across groups (box/median/whiskers/outliers). Label everything — axes, title, legend — every time.

**Quick checks 15.** (a) What are the five steps of the plotting workflow? (b) Which plot type for a distribution of patient ages? (c) What does `alpha=0.5` do on a scatter plot? (d) In a box plot, what does the line inside the box represent? (e) What function displays the legend?

# Chapter 16 — Pharmaceutical Plots (Syllabus Point)

## 16.1 Concentration-time curves — oral vs IV administration

The **concentration-time curve** charts how plasma drug levels change over time and anchors all pharmacokinetic analysis. The two administration routes produce characteristically different shapes:

- **IV administration** — delivery is straight into the vein, so the trace **opens at maximum concentration** and falls steadily thereafter (elimination alone).
- **Oral administration** — the medicine must first dissolve and cross the gut wall, so the trace **climbs through an absorption phase**, summits at Cmax (reached at Tmax), then tapers under elimination.

The following example uses simple one-compartment pharmacokinetic equations to generate realistic curves:

```
import numpy as np
import matplotlib.pyplot as plt

t = np.linspace(0, 12, 100)       # 0 to 12 hours, 100 points
ka, ke = 1.2, 0.3                  # absorption & elimination rate constants

iv_conc = 45 * np.exp(-ke * t)                                    # IV:
elimination only
oral_conc = 60 * (np.exp(-ke * t) - np.exp(-ka * t)) * ka / (ka - ke)  # Oral:
absorption + elimination

plt.plot(t, iv_conc, color="teal", linewidth=2, label="IV Administration")
plt.plot(t, oral_conc, color="orange", linewidth=2, label="Oral Administration")
plt.xlabel("Time (hours)")
plt.ylabel("Plasma Concentration (mcg/mL)")
plt.title("Concentration-Time Curve: Oral vs IV")
plt.legend(loc="upper right")
plt.grid(alpha=0.3)
plt.show()
```

**Reading these curves:**

- The IV trace (teal) opens at its maximum and falls continuously — no absorption phase.
- The oral trace (orange) climbs from zero to a summit (Cmax at Tmax), then tapers under elimination.
- The **area beneath each trace (AUC)** measures cumulative drug exposure — a key bioequivalence parameter.

## 16.2 ADR reporting rates — the bar chart

A **bar chart** compares a count or rate across categories — the standard view for ADR reporting by drug:

```
import matplotlib.pyplot as plt

drugs = ["Metformin", "Amoxicillin", "Ibuprofen", "Atorvastatin", "Amlodipine"]
adr_counts = [22, 17, 11, 9, 6]

plt.bar(drugs, adr_counts, color=["teal", "orange", "teal", "orange", "teal"])
plt.xlabel("Drug Name")
plt.ylabel("Number of ADR Reports")
plt.title("ADR Reporting Rates Across Drugs")
plt.xticks(rotation=30)
plt.show()
```

`plt.xticks(rotation=30)` tilts the drug names so they do not overlap — a common fix for categorical labels.

**Interpretation caution:** Metformin tops this chart, though a count alone does not show *why* — heavier prescribing may account for it rather than a genuinely riskier profile. Weigh the denominator (how many receive each drug) before ranking risk.

## 16.3 Dissolution profiles — comparing formulations

A **dissolution profile** tracks the fraction of drug released over time — the standard comparison for release behaviour differences between tablet formulations, indispensable in quality assurance and bioequivalence work.

```
import matplotlib.pyplot as plt

time_min = [0, 5, 10, 15, 30, 45, 60, 90]
form_a = [0, 22, 38, 52, 74, 86, 93, 98]   # fast-release
form_b = [0, 10, 18, 26, 45, 58, 68, 80]   # sustained-release

plt.plot(time_min, form_a, color="teal", marker="o", label="Formulation A (fast-
release)")
plt.plot(time_min, form_b, color="orange", marker="s", label="Formulation B
(sustained-release)")
plt.xlabel("Time (minutes)")
plt.ylabel("% Drug Dissolved")
plt.title("Dissolution Profile Comparison")
plt.legend(loc="lower right")
plt.grid(alpha=0.3)
plt.show()
```

**Reading the dissolution plot:** Formulation A reaches 85% dissolved within 30 minutes (fast-release behaviour); Formulation B takes about 60 minutes (sustained-release). The two curves diverge most between 15 and 45 minutes — the window where formulation choice matters clinically. Similar

shapes and AUCs may suggest bioequivalence; clearly different profiles mean the formulations are not interchangeable.

## 16.4 The mini PK study — a complete workflow

The capstone of the book: load a concentration CSV → clean with Pandas → compute parameters with NumPy → plot with Matplotlib. Every unit's skills in one program:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

# 1. Read the PK dataset
pk = pd.DataFrame({
    "Time_h": [0, 0.5, 1, 2, 4, 6, 8, 10, 12],
    "Conc": [0.0, 8.2, 15.6, 22.1, 18.3, 12.5, 8.1, 5.2, 3.1],
})

# 2. Compute pharmacokinetic parameters
cmax = pk["Conc"].max()
tmax = pk.loc[pk["Conc"].idxmax(), "Time_h"]
auc = np.trapezoid(pk["Conc"], pk["Time_h"])

# 3. Plot the curve
plt.plot(pk["Time_h"], pk["Conc"], color="teal", marker="o", linewidth=2)
plt.axhline(cmax, color="orange", linestyle="--", alpha=0.7, label=f"Cmax =
{cmax:.1f}")
plt.axvline(tmax, color="gray", linestyle=":", alpha=0.7, label=f"Tmax =
{tmax:.1f} h")
plt.fill_between(pk["Time_h"], pk["Conc"], alpha=0.15, color="teal", label=f"AUC
= {auc:.1f}")
plt.xlabel("Time (hours)")
plt.ylabel("Plasma Concentration (mcg/mL)")
plt.title("Mini PK Study: Concentration-Time Curve")
plt.legend(loc="upper right")
plt.grid(alpha=0.3)
plt.show()

print(f"Cmax = {cmax:.1f} mcg/mL at Tmax = {tmax:.1f} h")
print(f"AUC (0-12h) = {auc:.1f} mcg*h/mL")
```

OUTPUT

```
Cmax = 22.1 mcg/mL at Tmax = 2.0 h
AUC (0-12h) = 140.2 mcg*h/mL
```

Three units in one script: Pandas reads and structures; NumPy computes the AUC by the trapezoidal rule; Matplotlib draws the curve with Cmax (dashed line), Tmax (dotted line) and the AUC (shaded area) annotated. This is the shape of every real PK analysis — and it fits on one page.

**Chapter summary.** Concentration-time curves compare IV (immediate peak, pure elimination) with oral (absorption phase → Cmax at Tmax → elimination); the AUC reflects total exposure. ADR rates are compared with bar charts — but always contextualise counts against prescription volume. Dissolution profiles compare formulation release rates; separate curves signal non-interchangeability. The mini PK study (§16.4) integrates the entire book: Pandas for data, NumPy for computation, Matplotlib for the figure.

**Quick checks 16.** (a) Why does an IV curve start high and only decline? (b) What does the AUC represent? (c) Why rotate x-tick labels on a bar chart? (d) In the dissolution example, which formulation reaches 85% faster? (e) Name the three libraries used in the mini PK study and their roles.

# Chapter 17 — Scientific Interpretation of Plots (Syllabus Point)

## 17.1 The figure is only step one

Correctly reading and drawing conclusions from a chart is the skill that gives a figure its value in pharmaceutical work. This section provides the interpretation framework the syllabus expects.

## 17.2 The interpretation reference table

| Plot feature | What to look for | Pharmacy meaning |
|---|---|---|
| Peak height (Cmax) | higher or earlier peak | faster or more complete absorption |
| Curve steepness (decline) | steeper fall | faster elimination (higher clearance) |
| Overlapping curves | similar shape and AUC | may suggest bioequivalence between formulations |
| Box-plot spread (IQR) | broader box | greater manufacturing inconsistency — a quality concern |
| Outlier points | dots beyond whiskers | may indicate an assay error *or* a genuine unusual patient/batch |
| Scatter-plot trend | clear upward (or downward) pattern | suggests a dose-response correlation |

## 17.3 Correlation is not the same as causation

A scatter graph on which dose and response climb in tandem does **not**, on its own, demonstrate that the dose *produces* the response. A confounder such as disease severity, age, or a co-prescribed drug may influence both. Causation demands a controlled trial, not an observed association. This distinction matters daily in pharmacovigilance: an ADR cluster after a drug's launch may reflect increased prescribing, not increased risk.

## 17.4 Common interpretation mistakes

| Mistake | Why it matters | The fix |
|---|---|---|
| Reading a bar chart without context | highest count ≠ highest risk | always ask "how many patients were on each drug?" |
| Ignoring outliers | they may be the most important data points | investigate: assay error or genuine signal? |
| Over-reading small samples | 3 points cannot establish a trend | report sample size alongside every figure |

| Mistake | Why it matters | The fix |
|---|---|---|
| Confusing AUC with Cmax | AUC = total exposure; Cmax = peak level | both matter, but for different clinical questions |

**Chapter summary.** Every plot tells a story: peaks reveal absorption, steepness reveals elimination, overlap suggests bioequivalence, spread reveals manufacturing consistency, outliers warrant scrutiny. Correlation observed on a scatter plot is a starting point for study, not proof of cause. Always contextualise counts (prescription volume), investigate outliers, and report sample sizes.

**Quick checks 17.** (a) What does a steeper decline on a concentration-time curve suggest? (b) A wide box in a box plot means what? (c) Give one reason why correlation does not prove causation. (d) Why report sample size alongside every figure?

# Unit V — Consolidation

## Syllabus coverage map

| Syllabus bullet | Where |
|---|---|
| Introduction to Matplotlib | Ch 15 (§15.1–15.2) |
| Creating line plots, histograms, scatter plots, box plots | Ch 15 (§15.3–15.6) |
| Labeling axes, titles, legends | Ch 15 (§15.7) |
| Pharmaceutical plots: concentration-time, ADR rates, dissolution | Ch 16 (§16.1–16.4) |
| Scientific interpretation of plots | Ch 17 |

## Twenty-question exercise bank

*Short answers (1–8).*

1. Name the four core plot types and their use cases.
2. What are the five steps of the plotting workflow?
3. What does the `bins` parameter control in a histogram?
4. In a box plot: what do the box edges, the line inside, and the whiskers represent?
5. What does `alpha=0.5` do?
6. Why is a legend essential when overlaying two datasets?
7. How does an IV concentration-time curve differ from an oral one?
8. What does the AUC represent on a concentration-time curve?

*Interpretation (9–14).*

9. A box plot of Batch X is much wider than Batch Y. What does this mean?
10. A scatter plot of dose vs response shows a clear upward trend. Can you conclude that higher doses cause better response?
11. A bar chart shows Drug A with 100 ADR reports and Drug B with 20. Is Drug A riskier?
12. Two dissolution profiles have almost identical shapes. What might this suggest?
13. A concentration-time curve drops very steeply after the peak. What pharmacokinetic property does this reflect?
14. An outlier dot appears in a box plot of Cmax values. List two possible explanations.

*Write programs (15–20).*

15. Plot a line graph of temperature readings over 24 hours with proper labels.
16. Create a histogram of 20 patient ages with `bins=8`.

17. Scatter plot dose (x) against response (y) for 8 data points; add labels.
18. Box plot Cmax values for two batches; label the batches.
19. Plot an oral concentration-time curve (use the §16.1 model) with Cmax annotated.
20. Bar chart ADR counts for five drugs; rotate the labels.

## Answers to selected exercises

1. Line (time-series), histogram (distribution), scatter (relationship), box (five-number summary). 2. Plot → xlabel → ylabel → title → show. 3. The number of groups the data is divided into. 4. Q3 & Q1 (edges), median (line), typical range (whiskers). 5. Makes the plotted element 50% transparent. 7. IV starts at its peak and only declines; oral rises to a peak then declines. 8. Total drug exposure over the measured period. 9. Greater manufacturing inconsistency in Batch X. 10. No — association is not causation; confounders may exist. 11. Not necessarily — Drug A may simply have more patients. 12. Bioequivalence between the two formulations. 13. High clearance / rapid elimination. 14. Assay error; genuine unusual patient or batch.

## Important points for examination

1. Matplotlib is imported as `import matplotlib.pyplot as plt` — the universal alias.
2. The plotting workflow: choose plot type and pass data → label x-axis → label y-axis → add title → display.
3. Four core plot types: **line** (`plt.plot`) for time-series; **histogram** (`plt.hist`) for frequency distributions; **scatter** (`plt.scatter`) for relationships between two variables; **box plot** (`plt.boxplot`) for five-number summaries (median, quartiles, whiskers, outliers) across groups.
4. Customisation parameters: `color`, `marker`, `linewidth`/`lw`, `linestyle`/`ls`, `alpha`, `label` (for the legend), `s` (scatter size), `bins` (histogram groups), `edgecolor` (histogram bar borders).
5. Every scientific figure must be fully labelled: `xlabel`, `ylabel`, `title`, and `legend()` whenever more than one dataset appears.
6. **Concentration-time curves**: IV administration shows an instant maximum followed by elimination alone; oral administration shows absorption → Cmax at Tmax → elimination; the AUC measures cumulative drug exposure.
7. **ADR rates** are compared with bar charts (`plt.bar`); `xticks(rotation=...)` prevents label overlap; weigh counts against how widely each drug is prescribed.
8. **Dissolution profiles** compare formulation release rates; matching curves hint at bioequivalence; separate curves signal non-interchangeability.
9. **Interpretation**: steeper fall = quicker clearance; broader box = more variability; overlapping curves = possible bioequivalence; outliers warrant scrutiny.

10. **Correlation is not the same as causation** — an association on a scatter plot is a lead to investigate, not a finding to publish as proven.

*Sources for this unit:* course materials; topic coverage cross-checked against Noteskarts *Computer Applications in Pharmacy — Unit 5* notes, matplotlib.org documentation and gallery, MCA R24 lecture notes, and the curated learn.biodockify.com catalog (matplotlib, numpy, pandas). All prose and examples are original; code outputs verified by execution; originality audited against every consulted source. Full credits: see the book's Sources and Acknowledgements page.